

Faculty of Environmental Sciences

Hydro Science and Engineering

Impact of Decentralized Rainwater Management in the Räcknitz District (TUD Campus)

Report on

Urban Water II – Modelling of Wastewater Systems (MHSE22)

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Abstract

Urban development often increases impervious surfaces, which alters the natural hydrological cycle and increases surface runoff in urban drainage systems. Decentralized stormwater management strategies, particularly Low Impact Development (LID) measures, aim to mitigate these impacts by promoting infiltration, evapotranspiration, and temporary water storage. This study evaluates the impact of decentralized rainwater management measures on the water balance of the Räcknitz district in Dresden using the Storm Water Management Model (SWMM).

A hydrological model of the study area was developed and calibrated using observed flow data from selected nodes in the sewer network. Model performance was evaluated using the Nash–Sutcliffe Efficiency (NSE) and Peak Flow Error (PFE) indicators to ensure accurate representation of the drainage system. In addition, a new development area representing an extension of the university campus was incorporated into the model to assess its influence on the hydraulic loading of the existing sewer network.

Three stormwater management scenarios were analyzed, including Short-term LID, Long-term LID, and a Combined LID scenario. The performance of each scenario was evaluated based on outfall loading, peak discharge, flooding conditions, and long-term water balance simulations covering the period from 1996 to 2015.

The results indicate that while the new development area slightly increases runoff entering the sewer network, the implementation of LID strategies significantly reduces surface runoff and peak discharge. The Combined LID scenario produced the greatest reduction in runoff and improved overall system performance by enhancing infiltration and evapotranspiration processes. These findings demonstrate the effectiveness of decentralized stormwater management approaches in improving urban water balance and reducing hydraulic stress on existing drainage infrastructure.

Keywords: Decentralized stormwater management, LID, SWMM, Urban water balance, NDA, NSE, PFE, Runoff Reduction, Räcknitz district, Hydrological Modelling.

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List of Abbreviations

DWF	Dry Weather Flow
EPA	Environmental Protection Agency
EPSG	European Petroleum Survey Group
GOF	Goodness of Fit
IDF	Intensity-Duration-Frequency
LID	Low Impact Development
NDA	New Development Area
NSE	Nash–Sutcliffe Efficiency
PFE	Peak Flow Error
SWMM	Storm Water Management Model

1 Introduction

1.1 Background

Urbanization has transformed natural landscapes into densely built environments dominated by impervious surfaces such as roads, rooftops, and parking areas. These changes disrupt the hydrological cycle by reducing infiltration, limiting groundwater recharge, and increasing surface runoff. Consequently, urban catchments often experience higher peak discharges, faster runoff responses, and a greater risk of flooding. Expanding urban areas also place pressure on wastewater and drainage infrastructure, frequently causing stormwater overflows, waterlogging, and degraded water quality in receiving water bodies (Zoppou, 2001). Combined with rapid urban growth and the increasing impacts of climate change; particularly more frequent and intense rainfall events effective stormwater management has become a critical challenge for sustainable urban development (Ashley et al., 2005).

Traditional stormwater management relies on centralized drainage networks that rapidly convey runoff to rivers or treatment facilities. While effective at reducing localized flooding, these systems often fail to address the underlying causes of increased runoff, limit infiltration, restrict groundwater recharge, and may exacerbate downstream flooding. Many existing systems were designed based on historical rainfall and are increasingly inadequate under changing climatic and urban conditions (Zoppou, 2001; Ashley et al., 2005).

Sustainable, decentralized approaches such as Low Impact Development (LID) replicate natural hydrological processes through infiltration, evapotranspiration, temporary storage, and gradual runoff release. Typical LID measures; green roofs, bio-retention cells, rain gardens, permeable pavements, infiltration trenches, and vegetated swales—reduce runoff, improve water quality, and support groundwater recharge (Dietz, 2007; Eckart et al., 2018; Fletcher et al., 2015).

Hydrological models, particularly the EPA Storm Water Management Model (SWMM), simulate rainfall–runoff dynamics, sewer network performance, and pollutant transport, allowing evaluation of alternative stormwater management strategies (Rossman, 2010). In this study, SWMM is applied to the Räcknitz district of Dresden to assess the effectiveness of short- and long-term LID measures and the impact of a new development area on runoff and overall water balance.

1.2 Objectives

The main objectives of this study are summarized as follows:

- Simulate the existing water balance and runoff conditions of the Räcknitz district using SWMM.
- Evaluate the impact of a new development area on the capacity of the existing sewer network.
- Assess Low Impact Development (LID) measures for decentralized storm-water management and runoff reduction.
- Compare short-term, long-term, and combined LID scenarios for improving the water balance and reducing sewer overflow risk according to DWA-A 118.

1.3 Study Area

The study area is in the Räcknitz district in Dresden, Germany. It is an urban catchment with a mix of residential, institutional, traffic, and open space areas. For the modelling, the catchment was divided into 19 sub-catchments with a total area of about 25 ha. The rainfall data used for the simulation was taken from a nearby rain gauge and provided in 5-minute intervals.

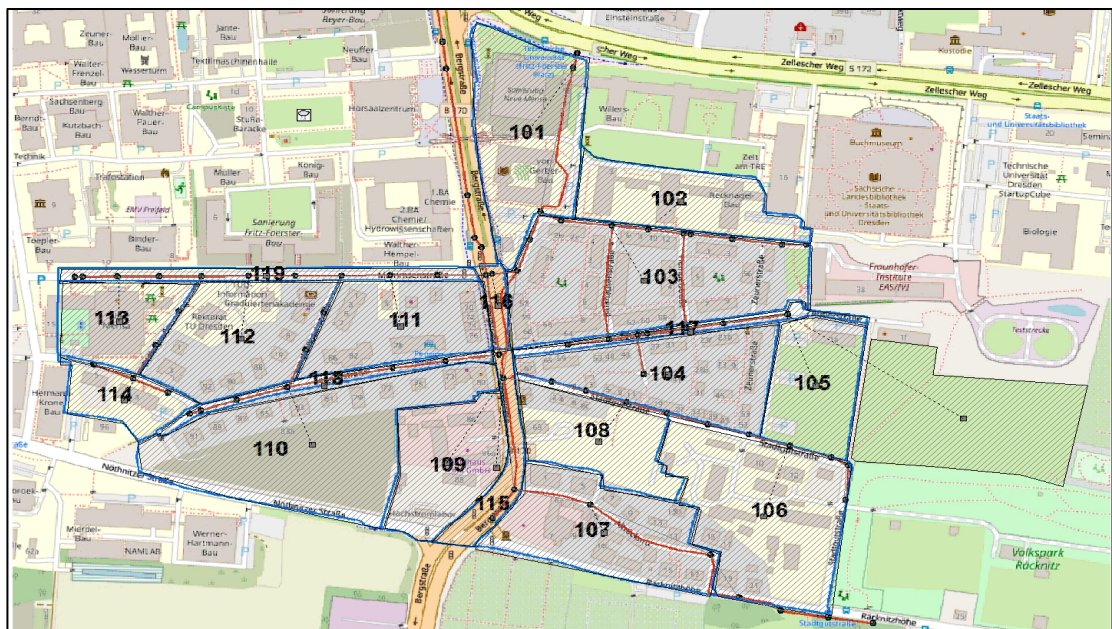


Figure 1: Räcknitz District Catchment in Dresden

In addition to the existing catchment, the study also considers a planned development area related to the extension of the university campus. This site is located east

of the seminar buildings on Zeunerstraße and south of Volkspark Räcknitz. The area covers about 3 ha and, according to the task description, includes 6,000 m² of building area, 2,500 m² of traffic area, and around 100 population equivalents.

2 Materials and Methods

2.1 Modelling Framework and software

This study aims to evaluate the impact of decentralized rainwater management on the long-term water balance of the Räcknitz district in Dresden. Continuous hydrological simulations were conducted using EPA SWMM version 5.2, which represents surface runoff generation, flow routing through sewer networks, and interactions between hydrological and hydraulic processes under dry and wet conditions. The model was run in an event-based configuration to assess system responses to selected rainfall events and compare alternative stormwater management scenarios. Hydraulic routing used the dynamic wave method, accounting for pressurized flow, backwater effects, and flow reversals, with a minimum conduit slope of 1 % and no surface ponding to ensure stability and realistic results.

2.2 Model Data

2.2.1 Sewer Network Modelling

The sewer network was implemented in SWMM using available spatial and hydraulic data (Annex III) including conduits and junctions (Annex III-G, Annex III-H) with attributes such as pipe length, diameter, invert elevation, roughness, and manhole depth. For consistency and to reduce parameter uncertainty, all conduits were assumed circular and made of concrete, allowing a uniform Manning's roughness coefficient. Network elements were georeferenced using EPSG 3857, ensuring accurate spatial integration with sub-catchments. This approach provides a hydraulically coherent representation of the existing sewer system while maintaining computational efficiency.

2.2.2 Dry Weather Flow

Dry Weather Flow (DWF) was included to account for wastewater contributions under rainfall-free conditions. In this study, DWF consists primarily of domestic wastewater generated within the catchment, while industrial effluents and

groundwater infiltration were considered negligible, simplifying total DWF to domestic sewage flow only.

The total dry weather flow (Q_{dw}) is calculated using the following equations:

$$Q_{dw} = Q_s + Q_e$$

Equation 1: Dry weather flow Equation

Where,

$$Q_s = Q_{dom} + Q_{ind}$$

Q_{dw} = Dry weather flow (DWF)

Q_s = Sewage flow

Q_e = Extraneous flow (typically 30% to 50% of Q_s)

Q_{dom} = Domestic sewage flow

Q_{ind} = Industrial sewage flow

In our case, $Q_s = Q_{dom}$ (since $Q_{ind} = 0$)

Therefore, $Q_{dw} = Q_{dom}$.

Temporal variability in DWF was represented using diurnal flow patterns reflecting typical household water consumption. Separate profiles were applied for weekdays and weekends to capture usage differences (Annex III-E). This approach ensures realistic base flow conditions and improves the reliability of wet weather flow simulations. Annex III-B provides an overview of the water consumption values for the study area.

2.2.3 Sub-Catchment Delineation and Hydrological Parameters

The study area was subdivided into multiple sub-catchments, each representing a hydrologically homogeneous unit connected to the sewer network. Sub-catchment boundaries were defined using cadastral data and land-use information, ensuring an accurate spatial representation of surface drainage pathways.

Each sub-catchment was characterized by key hydrological parameters, including area, average slope, characteristic width, and percentage of impervious surfaces whose values are presented in Annex III-C. Imperviousness was derived from land-use classification, distinguishing paved areas such as roads, sidewalks, and flat roofs from pervious surfaces including green spaces, gardens, and pitched roofs.

These parameters govern runoff generation and conveyance to the sewer system and were carefully selected to reflect the physical characteristics of the study area Annex III-A: Sub catchment Delineation Data.

2.3 Calibration and Validation

2.3.1 Classification of Rain Events

Rainfall events were classified into three intensity levels based on their duration and return period, as presented in Table 1. This classification helps represent different rainfall conditions that may influence the hydrological response of the system. The events were grouped into short, medium, and high categories to capture a range of storm characteristics.

Table 1: Classification of rain events based on duration and return period

Parameter	Level		
	Short	Medium	High
Duration (min)	30-120	120-240	>240
Return Period (Year)	0.5-2	02-10	>10

2.3.2 Selection of Rain Events and calibration

After classification, these events were identified from the available precipitation time series covering the period from January 1, 1996, to November 30, 2015 (Rain.CO.dat file). Out of a total of 47 potential rainfall events, at least six were selected for the calibration process (*J.D. Reyes-Silva, Institute of Urban Water Management, TU Dresden*). The selected events, presented in Table 2, include key characteristics such as total rainfall, event duration, rainfall intensity, and return period, allowing representation of different storm conditions. The rainfall data were obtained from the Rain_CO.dat file, which contains precipitation records at five-minute intervals. The data were processed using Microsoft Excel to calculate the total rainfall amounts and the duration of each event. The return period for each occurrence was computed using Equation 1, and rainfall intensity was calculated using Equation 2.

$$z = \left(\frac{r_{tN(z)}}{r_{15(1)}} * \frac{t_N + 9(\text{min})}{38(\text{min})} + 0.369 \right)^4$$

Equation 2: Return Period

$$r_{t_R,z} = r_{15(1)} * \frac{38(\text{min})}{t_n + 9(\text{min})} + \left(z^{\frac{1}{4}} - 0.369 \right)$$

*Equation 3: Reinhold's Intensity-Duration-Frequency
(IDF) Equation (Rainfall Intensity)*

Where:

z = Return Period of the rain event

$r_{t_N(z)}$ = Rainfall intensity as a function of rain duration t_n and return period z

$r_{15(1)}$ = Rainfall intensity of 15 min duration and 1 year return period (116.7 L/s*Ha for Dresden)

t_N = Duration of Rainfall in minutes

The rainfall events listed in Table 3 were used to successfully calibrate the SWMM model.

Table 2: Rain event details evaluated during Calibration

Calibration						
Rain event (D/M/Y)	Dura- tion (min)	Level	Total Rainfall (mm)	Intensity (mm/hr)	Return period (years)	Level
22/08/1996	50	Short	4.1	4.92	0.092	Short
09/07/1999	115	Short	0.5	0.26	0.023	Short
21/07/1998	145	Me- dium	23.8	9.85	3.046	Me- dium
08/08/2008	325	Me- dium	37.5	6.92	10.96	High
30/11/2002	1320	High	25.3	1.15	3.11	Me- dium
17/08/2015	800	High	50.6	3.79	27.652	High

After selecting representative rainfall events, slope, imperviousness, and width were adjusted within a predefined range (Annex III-B: Calibration Sub catchment wise Ranges) to improve model accuracy The specified values are shown in Table 6.

2.3.3 Performance Indicators

The accuracy of the SWMM model calibration was evaluated using two performance indicators: Nash–Sutcliffe Efficiency (NSE) and Peak Flow Error (PFE) in Equation 4 and Equation 5. These indicators assess how well the simulated results match the observed flow data.

NSE measures the Goodness of Fit (GOF) by comparing the variance of observed data with the squared differences between observed and simulated flows. Values close to 1 indicate strong agreement and ideal calibration (*J.D. Reyes-Silva, Institute of Urban Water Management, TU Dresden*).

$$NSE = 1 - \frac{\sum_{i=1}^n (Y_i^{obs} - Y_i^{sim})^2}{\sum_{i=1}^n (Y_i^{obs} - \text{mean}(Y^{obs}))^2}$$

Equation 4: Nash-Sutcliffe Efficiency

PFE evaluates the relative difference between peak observed and simulated flows, crucial for flood-related events, with PFE = 0 representing perfect agreement (*J.D. Reyes-Silva, Institute of Urban Water Management, TU Dresden*).

$$PFE = \left| \frac{\max(Y^{sim}) - \max(Y^{obs})}{\max(Y^{obs})} \right|$$

Equation 5: Peak Flow Error

Where:

i = Time step

n = Total amount of time steps

Y^{obs} = Time series of observed data

Y^{sim} = Time series of simulated data

Calibration involved adjusting model parameters to match observed flows at Nodes 06L95–06L98 and 06M5–06M4. NSE and PFE were calculated for each time step across six selected rainfall events, with data organized in Excel to support calibration. Model performance was deemed acceptable when NSE exceeded 0.85 and PFE remained below 0.2. Goodness-of-fit analysis demonstrated strong agreement between simulated and observed flows, confirming that the SWMM model met the required accuracy criteria (Table 4).

2.3.4 Validation

For model validation, the average values of slope, width, and imperviousness obtained from the six calibration cases were applied to all 19 sub-catchments. The

calibrated model was then tested using three additional rainfall events representing low, medium, and high return periods. This approach allowed the model's performance to be evaluated under different rainfall conditions.

Like the calibration process, validation involved comparing observed and simulated flow data to ensure that the selected parameters performed consistently across the study area assessed using the NSE and PFE at key nodes. The results, summarized in Table 3, demonstrate good agreement between observed and simulated values, confirming that the model is sufficiently reliable for further analysis.

Table 3: Rain event details evaluated during Validation

Validation						
Rain event (D/M/Y)	Duration (min)	Level	Rainfall (mm)	Intensity (mm/hr)	Return period (years)	Level
27/06/2006	105	Short	20.3	11.6	2.06	Short
18/05/2014	345	Medium	19.9	3.46	1.671	Short
08/07/1996	1385	High	43.2	1.87	16.108	High

2.4 New Development Area (NDA)

To evaluate the impact of urban expansion on the existing stormwater drainage system, a new development area (NDA) of approximately 3 hectares was incorporated into the SWMM model. The NDA, representing a planned extension of the TU Dresden campus east of the Zeunerstraße seminar buildings in the Räcknitz district, includes 6000 m² of buildings for about 100 population equivalent and 2500 m² for roads, sidewalks, and parking. The total study area, originally 25 hectares divided into 19 sub-catchments, was updated to include this additional impervious surface, simulating increased hydraulic loads from urban development.

The NDA was modeled as a new sub-catchment (Sub-catchment 120) with typical urban land-use characteristics. Its hydraulic properties include a flow width of 188 m, 28.3 % imperviousness, and an assumed slope of 1.5 %. DWF was estimated using the population equivalent and an average per-capita consumption of 90 L/day. Discharge from the NDA was routed to the nearest downstream junction (Junction 06M64), avoiding additional pumps or regulatory structures.

The system performance was assessed according to the guidelines of DWA-A 118. For this purpose, a 1-hour Euler Type II rainfall event, corresponding to a 5-year return period and derived from KOSTRA Atlas data (Grid 138196, Annex III-F), was applied. This analysis allowed the capacity of the existing drainage network to be evaluated under increased hydraulic load from the proposed expansion while ensuring compliance with the relevant urban drainage standards.

2.5 Low Impact Development (LID)

Low Impact Development (LID) strategies were applied to manage stormwater at its source, reducing hydraulic loads from the NDA and improving system resilience. Selection of LIDs considered sub-catchment characteristics such as imperviousness, slope, soil type, and feasibility. Both short- and long-term measures were modeled for high-runoff sub-catchments (e.g., 105, 120, and areas near the NDA), with parameters based on DWA-A 118 and EPA SWMM guidelines, adapted to local Dresden conditions.

LIDs were simulated under various scenarios, including combined short- and long-term implementations, to assess reductions in runoff (target: 15-25%), infiltration enhancement (3-5 mm/h), and flood volume decreases (up to 50%). The integrated LID strategy aimed to replicate natural hydrology, promoting infiltration, retention, and evapotranspiration while complying with urban drainage standards.

2.5.1 Short-terms LIDs

Short-term LIDs focus on immediate, cost-effective measures to reduce peak flows during intense rainfall and were prioritized for rapid implementation in the NDA and nearby highly impervious areas (Figures in Annex I). Two short-term measures were modeled:

- **Infiltration Trenches:** Underground trenches filled with porous media to enhance infiltration capacity and control stormwater, typically located along roadsides to promote groundwater recharge (Conley et al., 2020; Hager et al., 2019). Dimensions are detailed in Annex II-A: Infiltration trench.
- **Rain Gardens:** Rain gardens are shallow vegetated depressions designed to capture, filter, and infiltrate stormwater, thereby replicating natural hydrological processes to reduce flooding and improve water quality. Dimensions are being justified in Annex II-B: Rain Garden.

2.5.2 Long Term LIDs:

Long-term LID employs sustainable, nature-based approaches integrated into urban planning to enhance resilience to climate change (Figure, Annex I). Measures include:

- **Green Roofs:** Vegetated roofs that reduce runoff via evapotranspiration and storage within roof media (Berndtsson et al., 2006; Dietz, 2007; Liu et al., 2021). Dimensions are being justified in Annex II-C: Green Roof.
- **Bio-Retention Cells:** Bioretention cells operate by retaining captured stormwater for a period through their filter media thereby reducing runoff volumes (Liu & Fassman-Beck, 2017; Mangangka et al., 2015). Dimensions are being justified in Annex II-D: Bio-Retention Cells.
- **Permeable Pavements:** Porous surfaces that allow infiltration, reduce runoff, and recharge groundwater, used in parking lots and low-traffic areas (Ahiablame et al., 2013; Brunetti et al., 2016; Huang et al., 2016; Kuang et al., 2011). Dimensions are being justified in Annex II-E: Permeable Pavements.

2.5.3 Combined LIDs:

A combined implementation of short-term and long-term LID measures was also applied in the model. For example, in sub-catchment 103, a green roof, rain garden, and infiltration trench were implemented together to evaluate their integrated impact on stormwater management.

3 Result

3.1 Calibration and Validation results

The results confirm that for the nine rain events, the GOF criteria was met, ensuring that the model is successfully calibrated and validated.

3.1.1 Sample calculation

Following is the sample for the calculation that has been done throughout the model.

a) Input Data

- Total Rainfall (P): 37.5 mm
- Duration (t): 5.4 hours (324 minutes)

- Reference Intensity (r_{15}): 116.7 L/(s.ha)

b) Intensity Calculation

- Intensity (I) in mm/hr:

$$I = \frac{P}{t} = \frac{37.5 \text{ mm}}{5.4 \text{ hr}} = 6.94 \frac{\text{mm}}{\text{hr}}$$

- Intensity (r_{tn}) in L/(s*ha):

Using the conversion factor (1mm/hr = 2.7778 L/(s*ha)):

$$r_{tn} = 6.94 * 2.7778 = 19.28 \frac{\text{L}}{\text{s*ha}}$$

c) Return Period (z) Calculation:

Using the Reinhold formula form your calibration log:

$$z = \left(\frac{r_{tn}}{r_{15}} * \frac{t + 9}{38} + 0.369 \right)^4$$

Substitute the values (t=32 min):

$$z = \left(\frac{19.28}{116.7} * \frac{324 + 9}{38} + 0.369 \right)^4$$

$$z = (0.1562 * 8.763 + 0.369)^4$$

$$z \approx 10.91 \text{ Years}$$

d) Event Classification

- Duration class: Medium (Since 5.4 hours is between 2 and 6 hours).
- Return period class: (Since 10.91 years is greater than 10 years)

3.1.2 Calibrated GOF

Table 4 shows the corresponding average NSE and PFE values obtained for both the nodes 06L95-06L98 and 06M5-06M4 at respective date after the calibration process is completed.

Table 4: NSE & PFE Values for Calibrated rain events

Rain event (D/M/Y)	06L95-06L98		06M5-06M4	
	NSE	PFE	NSE	PFE
22/08/1996	0.924	0.0348815	0.974	0.0067321
09/07/1999	0.942	0.04251642	0.923	0.0086421
21/07/1998	0.994	0.02555822	0.996	0.04871372
08/08/2008	0.995	0.0736465	0.998	0.0256124
30/11/2002	0.962	0.01095559	0.955	0.06314114
17/08/2015	0.996	0.02872104	0.995	0.02955131

3.1.3 Validated GOF

Table 5 shows the different dates chosen for validation based on their return periods and the average NSE and PFE value for nodes *06L95-06L98* and *06M5-06M4*, ensuring that the chosen events represent a range of hydrological conditions

Table 5: GOF results for Validated rain events

Rain event (D/M/Y)	06L95-06L98		06M5-06M4	
	NSE	PFE	NSE	PFE
27/06/2006	0.996	0.0108841	0.996	0.02742451
18/05/2014	0.984	0.08716721	0.993	0.03887362
08/07/1996	0.987	0.02638972	0.979	0.05866004

3.1.4 All Parameter

Table 6 is the representative rainfall events, slope, imperviousness, and width that were adjusted within a predefined range to improve model accuracy.

Table 6: Parameters for Calibration

Parameters for Calibration			
Sub-catchment	Imperviousness (%)	Width (m)	Slope (%)
101	70	400	3
102	46	115	1.5
103	44	160	6.5
104	32	150	4
105	16.8	150	8
106	21.3	212	5.8
107	29	83	4.5
108	8	155	2.8
109	26	215	4.5
110	18	180	4.5
111	14.5	90	2.5
112	20	240	4.8
113	65	138	4.5
114	15.5	65	0.85
115	85	345	0.7
116	65	165	9.5
117	88.5	20	6.5
118	74	26	0.6
119	25	15	1.5

3.2 Impact of New Development Area

The addition of the new sub-catchment substantially increased total runoff, causing flooding at 12 distinct nodes as shown in Figure 2. While the impact was widespread, the severity varied across the system; seven of the affected nodes, for example: 06L121, 06L56, 06M47, experienced minor surcharging, while the 06M series showed higher hydraulic stress. Notably, node 06M39 recorded the highest flood volume at 0.106×10^6 liters, followed by node 06M64 at 0.08×10^6 liters and node 06M41 at 0.052×10^6 liters, indicating that existing infrastructure at these locations cannot accommodate peak flows generated by the new development's impervious surfaces.

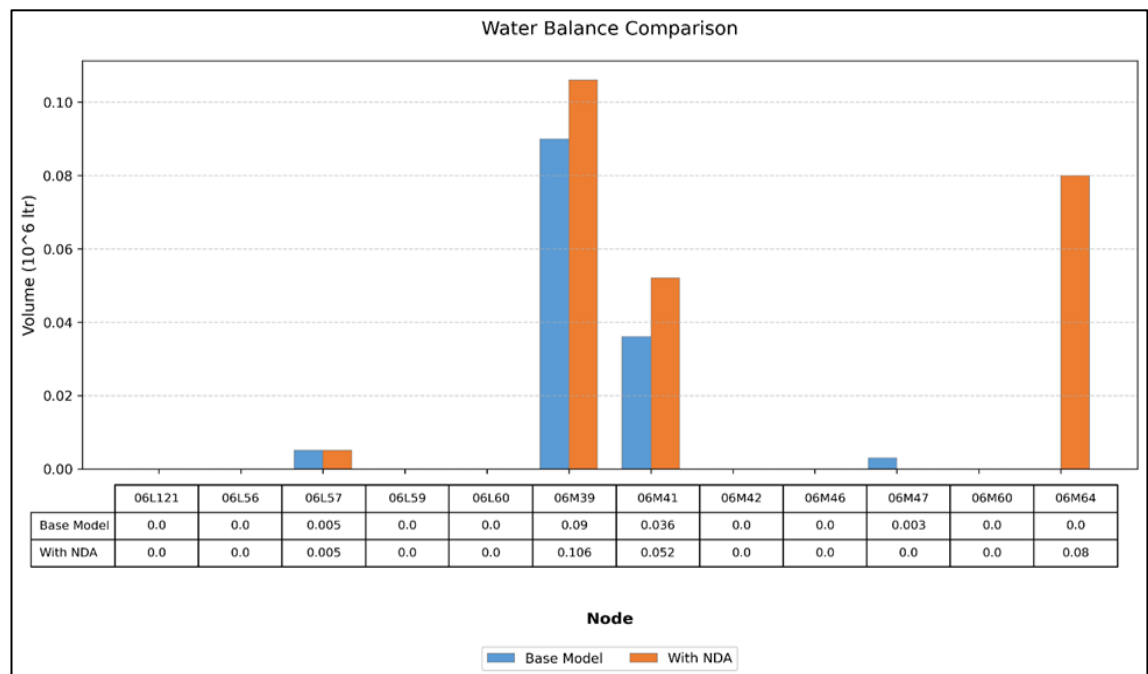


Figure 2: Comparison of water balance with base model and new development area

The comparison between the Base Model and the NDA as shown in Figure 3 indicates a clear shift in the hydrologic water balance due to increased urbanization. Surface runoff increased from 0.231 to 0.254 Ha-m due to added impervious area, infiltration rose from 0.431 to 0.488 Ha-m, likely from the larger catchment or soil characteristics defined for the new sub-catchment. Additionally, evaporation losses became more pronounced in the NDA model, reaching 0.015 Ha-m, while the final storage also showed a slight uptick. These combined changes confirm that while the new development has expanded the system's capacity to intercept water, it has simultaneously increased the volume of surface runoff that must be managed by the drainage network.

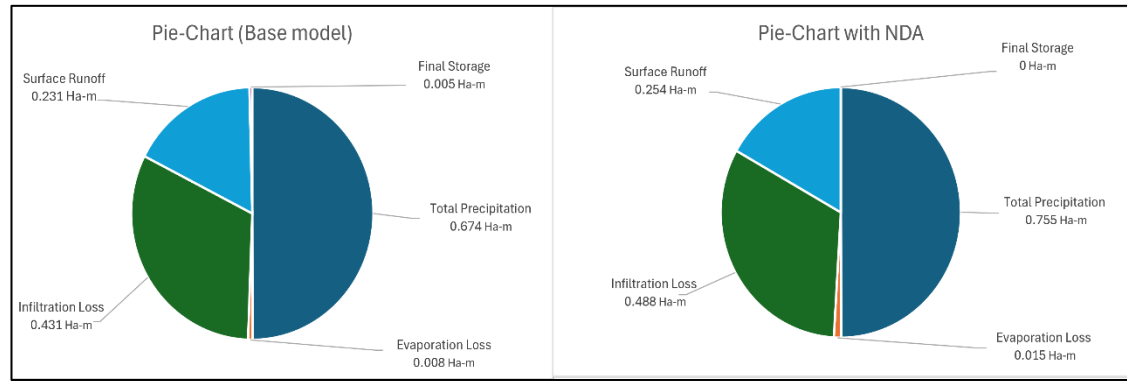


Figure 3: Pie-Chart showing water balance for base and NDA model for Euler's

3.3 Implementation of LIDs

To address hydraulic deficiencies, various LID measures were implemented to reduce flooding by managing runoff at its source through decentralized storage and infiltration, lowering peak flows to overloaded drainage infrastructure. LID features were placed in areas contributing to flooded nodes to evaluate their effectiveness in restoring system capacity and reducing local flood volumes.

LID measures were implemented as short-term, long-term, or combined strategies based on site-specific hydrological conditions and management objectives. Their selection considered catchment characteristics such as imperviousness, available area, soil type, and runoff potential. For example, green roofs were applied to large roof surfaces, and rain gardens to open spaces, with suitable locations identified using Google Maps satellite imagery to differentiate buildings from open areas. Table 7 gives the detailed categorization of LID that was been used in the sub-catchment area.

Table 7: Categorization of LIDs for each sub-catchment

Sub-catchment	Short-Term LIDs	Long-Term LIDs	Combined LIDs
S102	Infiltration Trench	Green Roof Bio Retention Cell Permeable pavement	Rain Garden Permeable pavement Infiltration Trench
S103	Rain Garden Infiltration Trench	Green Roof Bio Retention Cell	Green Roof Rain Garden Infiltration Trench

S104	Rain Garden Infiltration Trench	Green Roof Bio Retention Cell Permeable pavement	Green Roof Bio Retention Cell Infiltration Trench
S105	Rain Garden Infiltration Trench	Green Roof Bio Retention Cell Permeable pavement	Green Roof Rain Garden Permeable pavement
S112	Rain Garden	Bio Retention Cell	Infiltration Trench
S113	Rain Garden Infiltration Trench	Green Roof Bio Retention Cell	Rain Garden Green Roof
S117	Infiltration Trench	Permeable pavement	Infiltration Trench Permeable pavement
S120	Rain Garden	Bio Retention Cell Permeable pavement	Rain Garden Bio Retention Cell Infiltration Trench Permeable pavement

3.3.1 Example of Implemented LIDs

For example, in Figure 4 sub-catchment S113, a combined LID scenario of Rain Garden and Green Roof was implemented based on a detailed analysis of the catchment's surface characteristics. The total impervious area was determined manually using the EPA SWMM measurement tool and cross-referenced with Google Maps satellite imagery to ensure spatial accuracy. To manage this runoff effectively, a suite of Green Roofs and a Rain Garden were strategically selected, with their configurations designed to treat specific portions of the total impervious surface.

Specifically, Green Roof 1 consists of two units covering 540 m² with 50 m width, treating 28 % of the impervious area. While green Roof 2 also covers 520m² with a width of 50 m, managing 30 % of the impervious surface. Additionally, a 1010 m² Rain Garden with a width of 26 m was incorporated to manage an additional 26 % of the impervious runoff. Each LID's footprint and treatment percentage were calculated based on the specific impervious area directly connected to the unit to ensure optimal hydraulic efficiency across the sub-catchment.



Figure 4: Example of combined LID scenario implemented in sub-catchment 113

3.4 LID Scenario Comparison

A detailed overview of the data obtained from the different LID scenarios implemented in the study area is done to see the effectiveness of each of the scenario in the sewer network, comparing it to the original model (Model with NDA and LIDs implemented).

3.4.1 Outfall Loading Comparison

To assess the efficiency of LID measures in improving stormwater discharge in the Räcknitz district, outfall loading was examined at two principal outlets: 06L98 and 06M4. Figure 5 presents the outfall loading metrics for these scenarios.

At outfall node 06L98, the Base Model average flow of 87.34 LPS increased to 112.74 LPS with the addition of the NDA, reflecting the extra hydraulic load from the new development. Applying combined LID measures reduced this flow to 59.67 LPS, a total reduction of approximately 31.7 %. Similarly, at outfall node 06M4, the base flow of 279.47 LPS rose to 325.57 LPS under the NDA scenario. The combined LID scenario significantly lowered the flow to 168.40 LPS, representing nearly 40 %. These quantitative results demonstrate that while development initially increased runoff volumes, the strategic application of combined LID

features significantly outperformed both short-term and long-term individual scenarios in restoring hydraulic capacity at the primary outlets.



Figure 5: Average flow and total volume of outfall load at different LID scenarios

Total outfall volumes show similar trends. At 06M4, the volume decreased from 1.918×10^6 L to 1.353×10^6 L under combined LIDs, a reduction of 0.565×10^6 L (~29.5 %). The NDA raised total volume at both outlets: from 1.398 to 1.593×10^6 L at 06L98 and from 1.918 to 2.036×10^6 L at 06M4, reflecting the additional runoff generated by the development area. In contrast, outlet 06L98 shows a more modest but steady improvement, decreasing by 0.135×10^6 Ltr from its base level, which equates to a 9.7 % reduction.

Across the board, the transition from the Base Model to the Short-Term LID provides the largest initial "jump" in efficiency, while the Combined LID ultimately yields the maximum total decrement by integrating the benefits of both short-term and long-term interventions.

3.4.2 Node Flooding

After implementing the LID measures, all flooding within the drainage network was successfully eliminated. For this, we generate Euler Type II rainfall event for return period of 5 years and of 60-minute duration and it was selected according to DWA-A 118 by classification of our study area as urban-residential. The variation in water depth at nodes under different scenarios can be observed in Figure 6, which compares the Base Model, NDA, Short-Term LID, Long-Term LID, and Combined LID scenarios. The results indicate that the Combined LID scenario outperforms individual Short-Term or Long-Term measures in reducing water depths.

Selected nodes; 06L121, 06L56, 06L57, 06L59, 06L60, 06M39, 06M41, 06M42, 06M46, 06M47, 06M60, and 06M64, were analyzed due to their flooding under the Base Model and NDA scenarios and their relatively high-water depths. In

general, the NDA scenario increased water depths compared with the Base Model, reflecting the effect of additional impervious surfaces and intensified hydraulic loading. Critical nodes, such as 06M41 and 06M42, exhibited the highest depths. For instance, water depth at 06M41 rose from 0.93 m (Base Model) to 1.21 m under the NDA scenario, a 30.1 % increase, due to increased impervious surfaces from urban development.

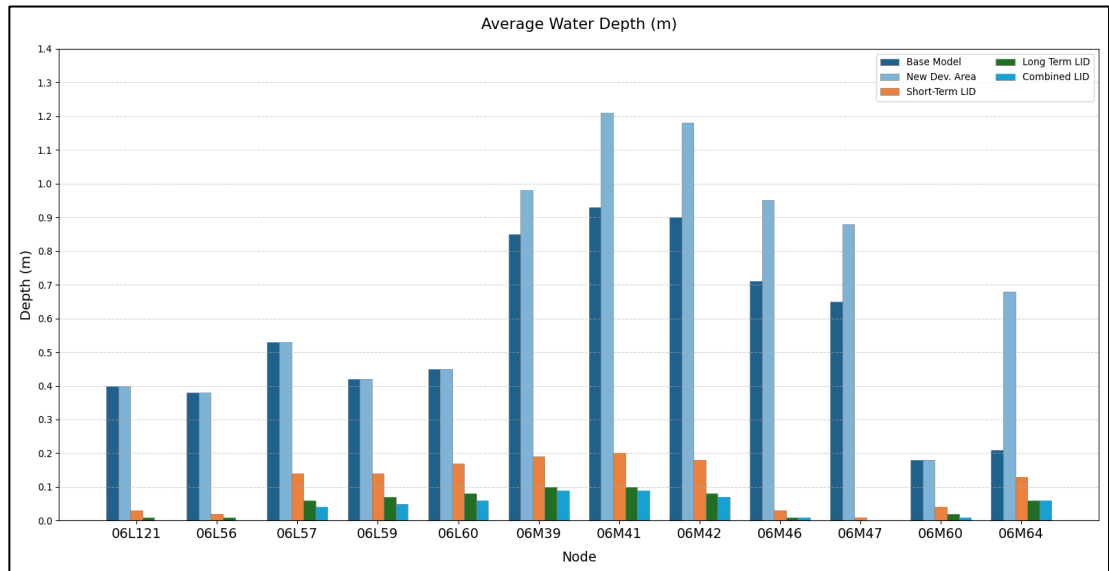


Figure 6: Average water depth at nodes for all LID scenarios

After implementing Short-Term LID measures, the depth decreased to approximately 0.20 m, corresponding to 83.6% reduction compared with the NDA scenario, while Long-Term LIDs further lowered it to 0.10 m (91.7 % reduction). The Combined LID scenario produced the greatest improvement, reducing the depth further to approximately 0.09 m, which corresponds to a 92.6 % reduction compared with the NDA scenario. Similar patterns were observed at other critical nodes, including 06M42, 06M64, and 06L60, where LID measures significantly mitigated the increased water depths caused by the NDA.

Across the network, the Short-Term LID scenario reduced water depths at most nodes compared with the NDA case, demonstrating the effectiveness of decentralized stormwater control measures in managing runoff. The Long-Term LID scenario also contributed to reducing nodal depths, although the reduction was slightly smaller at some nodes compared with the Short-Term LID scenario. Among all scenarios, the Combined LID strategy produced the most consistent reduction in water depth across the network. By integrating both short-term and long-term measures, this scenario more effectively controls runoff and improves the hydraulic performance of the drainage system.

3.4.3 Peak Discharge

The peak flow assessment was conducted at two key conduits, 06L95-06L98 and 06M5-06M4, to evaluate the impact of low-impact development (LID) strategies on stormwater dynamics. Figure 7 and Figure 8 illustrate the rise, peak, and recession of flow rates under multiple scenarios, including Base (pre-development), Short-Term LID, Long-Term LID, and Combined LID measures.

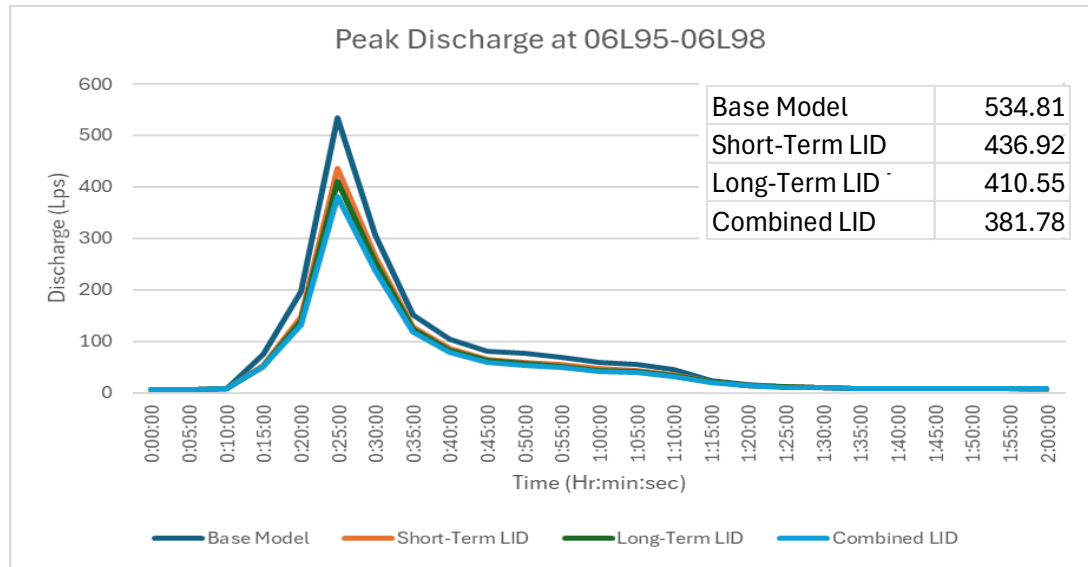


Figure 7: Peak Discharge in conduit 06L95-06L98 for all LID scenarios

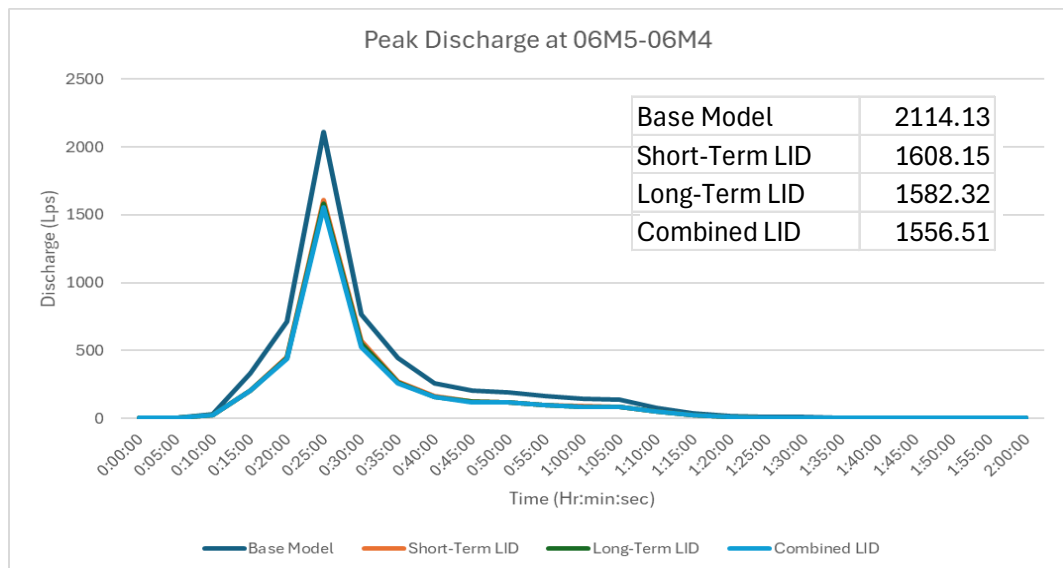


Figure 8: Peak Discharge in conduit 06M5-06M4 for all LID scenarios

At conduit 06L95-06L98, the Base scenario peak is 534.81 LPS at 00:25:00. Under LID measures, the peak flow decreases to 436.92 LPS with Short-Term LID (an 18.3% reduction), 410.55 LPS with Long-Term LID (a 23.2% reduction), and 381.78 LPS with Combined LID (a 28.6% reduction).

At conduit 06M5-06M4, the Base scenario shows a peak discharge of 2,114.13 LPS at 00:25:00. Application of LID strategies reduces the peak flows as follows: Short-Term LID decreases the peak to 1,608.15 LPS (a 23.9% reduction), Long-Term LID lowers it to 1,582.32 LPS (a 25.2% reduction), and the Combined LID scenario further reduces it to 1,556.51 LPS (a 26.4% reduction).

These results demonstrate that all LID strategies contribute to reducing peak flows, with the Combined LID scenario providing the most effective mitigation. Short-Term LID measures offer moderate reductions, while Long-Term LID techniques further enhance peak flow control. Combining both approaches maximizes the reduction at both conduits.

3.4.4 Water Balance Component

For the water balance assessment, the model was simulated for a long-term period from January 1, 1996, to November 30, 2015. This extended simulation allowed evaluation of cumulative precipitation effects on the catchment and provided a reliable representation of hydrological behavior under different management scenarios. The analysis quantified the distribution of precipitation among major water balance components over time.

Figure 9 illustrates water balance components under five scenarios: Base Model, NDA, Short-Term LID, Long-Term LID, and Combined LID. Precipitation is partitioned into surface runoff, infiltration loss, and evaporation loss. In the Base Model, infiltration dominates (7335.58 mm), followed by surface runoff (2584.23 mm) and evaporation (1163.11 mm), indicating that infiltration is the primary pathway for water loss under existing conditions.

In the NDA scenario, only minor changes are observed compared to the base model. Surface runoff slightly decreases to 2539.24 mm (1.74% reduction), infiltration increases to 7397.49 mm (0.84% increase), and evaporation remains nearly unchanged at 1146.00 mm, suggesting limited impact of the new development on overall water balance.

Short-Term LID implementation substantially reduces surface runoff to 1513.57 mm (41.4% reduction relative to the base model) and increases infiltration to 7796.67 mm (6.27% increase), reflecting improved infiltration from LID structures. Evaporation also rises to 1771.33 mm due to greater water retention and evapotranspiration. Long-Term LID further reduces runoff to 1469.17 mm (43.15% decrease) but shows a slight reduction in infiltration to 6918.83 mm

(5.67% decrease), while evaporation rises significantly to 2546.56 mm, highlighting enhanced evapotranspiration associated with vegetated LID systems. The Combined LID scenario achieves the greatest reduction in surface runoff, reaching 1424.21 mm (44.88% decrease), with infiltration loss remaining relatively high at 7403.37 mm (0.92% increase) and evaporation loss at 2088.17 mm.

Overall, the results demonstrate that the implementation of LID strategies significantly modifies the catchment water balance by reducing surface runoff and enhancing infiltration and evapotranspiration processes. Among the tested scenarios, the Combined LID configuration shows the highest effectiveness in reducing surface runoff, highlighting the potential of integrated LID approaches for sustainable stormwater management.

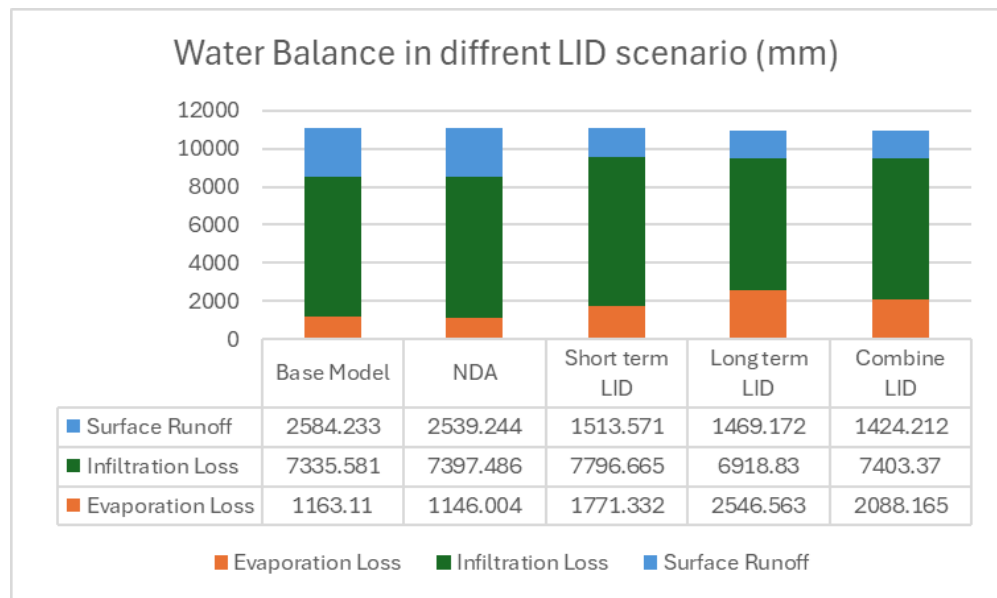


Figure 9: Water Balance of Study area in different LID scenarios

4 Discussion

The hydrological modelling results provide important insights into how urban development and decentralized stormwater management strategies affect the performance of urban drainage systems. The calibration and validation results indicate that the SWMM model reliably represents the hydrological response of the Räcknitz district catchment. High NSE values (~ 1.0) and low Peak PFE (~ 0) values demonstrate strong agreement between simulated and observed flows, confirming that the selected parameters, particularly sub-catchment width, slope, and imperviousness, appropriately represent the catchment characteristics.

The integration of the NDA clearly highlights the hydrological consequences of urban expansion. Increased impervious surfaces such as rooftops, roads, and parking areas resulted in higher surface runoff and peak flows within the sewer network. Consequently, more nodes experienced flooding, suggesting that the existing drainage infrastructure is already operating close to its hydraulic capacity. This behaviour is consistent with established urban hydrology principles, where increased imperviousness reduces infiltration and accelerates runoff generation, leading to higher peak discharge and increased flood risk.

The implementation of LID measures significantly improved the hydraulic performance of the system. By enhancing infiltration, evapotranspiration, and temporary storage, LIDs reduced runoff volumes and delayed peak flows entering the sewer network. Long-term measures such as green roofs, bio-retention cells, and permeable pavements provided sustained runoff reduction and improved the overall water balance of the catchment. In contrast, short-term measures including rain gardens and infiltration trenches offered flexible and cost-effective solutions for managing stormwater locally. The combined LID scenario showed the most balanced performance by integrating immediate runoff reduction with long-term hydrological benefits. This improved performance may also result from the flexibility to distribute different LID measures spatially across the catchment, allowing them to be implemented where they are most effective. However, the suitability and effectiveness of individual LID measures largely depend on local catchment characteristics such as land use, available space, and surface conditions.

Despite these positive results, some limitations must be considered. Hydrological modelling relies on assumptions regarding soil properties, infiltration rates, and land-use characteristics, which may vary spatially within the catchment. In addition, the simulations were based on selected rainfall events and may not fully represent the variability of extreme precipitation under future climate conditions. The simplified representation of LID performance in the model also does not fully capture real-world factors such as maintenance conditions or vegetation development.

Nevertheless, the results highlight the strong potential of decentralized stormwater management strategies to enhance the resilience of urban drainage systems. Integrating LID measures into urban planning can help mitigate the impacts of increasing urbanization while promoting more sustainable and climate-resilient water management.

5 Conclusion

This study evaluated the effects of decentralized rainwater management strategies on the stormwater system of the Räcknitz district in Dresden using the SWMM modelling framework. The hydrological model successfully simulated rainfall-runoff processes, providing a reliable basis to assess the impacts of urban development and alternative stormwater management scenarios.

Results indicate that the introduction of the new development area increases impervious surfaces, leading to higher runoff volumes, elevated peak flows, and additional hydraulic pressure on the existing sewer network. Several nodes experienced flooding under this scenario, suggesting that the current drainage infrastructure may have limited capacity to accommodate further urban expansion without supplementary management measures.

The implementation of LID measures effectively mitigated these impacts. Short-term measures, such as infiltration trenches and rain gardens, provided immediate, flexible runoff control, while long-term strategies, including green roofs, bio-retention cells, and permeable pavements, offered sustained reductions in runoff and improvements in the overall water balance. The combined LID scenario achieved the most effective performance by integrating both immediate and long-term benefits, reducing peak flows, limiting flooding duration, and enhancing infiltration and evapotranspiration.

These findings highlight the potential of decentralized stormwater management to restore natural hydrological processes, reduce hydraulic stress on sewer networks, and support compliance with urban drainage standards such as DWA-A 118. Incorporating LID strategies into urban planning can strengthen the resilience of drainage systems, mitigate flood risks, and promote long-term water balance stability. Future research should focus on optimizing spatial placement and combinations of LID measures, exploring additional solutions like rainwater harvesting, and evaluating performance under future climate variability.

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Appendix

Annex I: Figures of the LID used.



Infiltration Trench



Rain Garden



Green roofs



**Bio-Retention
Cells**



**Permeable
Pavements**

Annex II: Dimensions of LIDs used

Annex II-A: Infiltration trench

Referring to DWA-A-138,

Table 8: Parameters in LID control editor for Infiltration trench

Section	Parameter	Value
Surface (storage volume)	Berm Height (mm)	2500
	Vegetation Volume Fraction	0
	Surface Roughness (Mannings n)	0.1

	Surface Slope (%)	0
Storage (filter layer)	Thickness (mm)	500
	Void Ratio (Voids/Solids)	0.75
	Seepage Rate (mm/hr)	180
	Clogging Factor	0
Drain	Flow Coefficient	0
	Flow Exponent	1
	Offset (mm)	150
	Open Level (mm)	0
	Closed Level (mm)	0

Annex II-B: Rain Garden

Referring to Urban water modelling lectures (TU Dresden 2025),

Table 9: Parameters in LID control editor for Rain Garden

Section	Parameter	Value
Surface	Berm Height (mm)	110
	Vegetation Volume Fraction	0.15
	Surface Roughness (Mannings n)	0.1
	Surface Slope (%)	0.5
Soil	Thickness (mm)	300
	Porosity (volume fraction)	0.2
	Field Capacity (volume fraction)	0.1
	Wilting Point (volume fraction)	0.05
	Conductivity (mm/h)	37
	Conductivity Slope	10
	Suction Head (mm)	60
Storage	Thickness (mm)	0
	Void Ratio (Voids/Solids)	0.75
	Seepage Rate (mm/h)	10
	Clogging Factor	0

Annex II-C: Green Roof

Referring to DWA-A-102 and FLL Dachbegrünung,

Table 10: Parameters in LID control editor for Green Roof

Section	Parameter	Value
Surface	Berm Height (mm)	30
	Vegetation Volume Fraction	0.3
	Surface Roughness (Mannings n)	0.2
	Surface Slope (%)	10
Storage	Thickness (mm)	350
	Porosity (volume fraction)	0.4
	Field Capacity (volume fraction)	0.2
	Wilting Point (volume fraction)	0.05
	Conductivity (mm/hr)	180
	Conductivity Slope	30
	Suction Head (mm)	100
Drainage	Thickness (mm)	50
	Void Fraction	0.8
	Roughness (Mannings n)	0.1

Annex II-D: Bio-Retention Cells

Referring to DWA-A 138, AG Boden and RGL 601 Berlin,

Table 11: Parameters in LID control editor for Bio-Retention Cells

Section	Parameter	Value
Surface	Berm Height (mm)	300
	Vegetation Volume Fraction	0.08
	Surface Roughness (Mannings n)	0.15
	Surface Slope (%)	0.8
Soil	Thickness (mm)	200
	Porosity (volume fraction)	0.4

	Field Capacity (volume fraction)	0.2
	Wilting Point (volume fraction)	0.1
	Conductivity (mm/h)	37
	Conductivity Slope	20
	Suction Head (mm)	60
Storage	Thickness (mm)	170
	Void Ratio (Voids/Solids)	0.75
	Seepage Rate (mm/h)	20
	Clogging Factor	0
Drain	Flow Coefficient	0
	Flow Exponent	1
	Offset (mm)	110
	Open Level (mm)	0
	Closed Level (mm)	0

Annex II-E: Permeable Pavements

Referring to Urban water modelling lectures (TU Dresden 2025),

Table 12: Parameters in LID control editor for Permeable Pavements

Section	Parameter	Value
Surface	Berm Height (mm)	15
	Vegetation Volume Fraction	0.5
	Surface Roughness (Manning n)	0.15
	Surface Slope (%)	2
Pavement	Thickness (mm)	500
	Void ratio (voids/solids)	0.2
	Impervious Surface Fraction	0.5
	Permeability (mm/hr)	100
	Clogging Factor	0
	Regeneration Intervals (days)	0
	Regeneration Fraction	0
Storage	Thickness (mm)	150

	Void ratio (Voids/Solids)	0.25
	Seepage Rate (mm/hr)	150
	Clogging Factor	0
Drain	Flow Coefficient	0
	Flow Exponent	1
	Offset (mm)	150
	Open Level (mm)	0
	Closed Level (mm)	0
	Control Curve	0
Soil	Thickness (mm)	200
	Porosity (volume fraction)	0.4
	Field Capacity (volume fraction)	0.2
	Wilting Point (volume fraction)	0.1
	Conductivity (mm/h)	37
	Conductivity Slope	20
	Suction Head (mm)	60

Annex III: Data used for model setup

Table 13: Data used for model setup

Data Used	Description / Details
Precipitation (mm/5 min)	1996 to 2015
Evaporation (mm/d)	January to December (monthly)
Sub-catchment data	Area (ha)
	Impervious Area (ha)
Water consumption (m³/day)	Every street
Dry Weather Flow (m³/hour)	Weekdays
	Weekend
Conduit data	Material
	Shape
	Length (m)
	Diameter (mm)

Junction Data	Conduits in connection
	Co-ordinates (latitude, longitude)
	Invert elevation (m.a.s.l)
	Manhole depth (m)

Annex III-A: Sub catchment Delineation Data

Table 14: Sub catchment Delineation Data

Parameter	Value
Roughness of impervious surfaces (N-Imperv)	0.013
Roughness of pervious surfaces (N-Perv)	0.15
Depression storage of impervious surfaces (Dstore-Imperv)	1.27 mm
Depression storage of pervious surfaces (Dstore-Perv)	2.54 mm

Annex III-B: Water Consumption data of the study area

Table 15: Water Consumption data of the study area (Reyes-Silva, Water consumption, 2024)

Location	Water Consumption (m ³ /day)
Haeckelstraße	9,425
Zeunerstraße (East of Bergstraße)	32,424
Zeunerstraße (West of Bergstraße)	128,833
Bergstraße	462,328
Mommensenstraße	3,516

Annex III-C: Calibration Sub catchment wise Ranges

Table 16: Sub catchment Ranges for Calibration

Sub-catchment	Imperviousness (%)		Width (m)		Slope (%)	
101	33	100	190	570	1	4
102	20	65	56	167	1	2
103	18	57	70	209	3	8
104	17	54	77	230	2	5

105	7	22	71	212	3	10
106	11	36	102	306	2	7
107	14	46	39	116	2	5
108	4	12	72	216	1	4
109	13	42	102	306	2	5
110	9	28	88	263	2	6
111	7	23	43	129	1	3
112	9	30	117	351	2	6
113	32	100	67	201	2	6
114	8	26	30	89	0	1
115	45	100	157	470	0	1
116	32	100	75	225	4	11
117	45	100	9	27	3	9
118	38	100	12	35	0	1
119	14	46	6	18	1	2

Annex III-D: Evaporation data

Table 17: Evaporation data

Month	Evaporation (mm/day)
January	0.47
February	0.82
March	1.42
April	3.14
May	4.77
June	5.13
July	5.14
August	4.78
September	3.15
October	2.44
November	1.81
December	0.46

Annex III-E: Dry Weather Flow Pattern Data*Table 18: Dry Weather Flow Pattern Data*

Hour of day	Hourly_pattern_weekdays	Hourly_pattern_weekend
0:00	0.7549	0.8787
1:00	0.6616	0.825
2:00	0.5668	0.759
3:00	0.5197	0.6152
4:00	0.5465	0.498
5:00	0.7362	0.5831
6:00	1.0463	0.8693
7:00	1.2694	1.0798
8:00	1.3027	1.2212
9:00	1.2621	1.3105
10:00	1.2282	1.3155
11:00	1.2061	1.2144
12:00	1.191	1.1366
13:00	1.1667	1.0956
14:00	1.1358	1.0364
15:00	1.1061	1.0023
16:00	1.0817	1.0265
17:00	1.0646	1.1003
18:00	1.0554	1.1665
19:00	1.0497	1.1632
20:00	1.0612	1.1032
21:00	1.0822	1.0502
22:00	1.0236	1.0067
23:00	0.8815	0.9427

Annex III-F: KOSTRA Atlas Data*Table 19: 5-year return period; 1-hour Euler Type II storm pattern*

Year	Month	Day	Hour	Minute	Rainfall (mm)
2018	7	20	12	0	0

2018	7	20	12	5	1.8
0:00	7	20	12	10	2.4
2018	7	20	12	15	3.9
2018	7	20	12	20	11.5
2018	7	20	12	25	1.3
2018	7	20	12	30	1.3
2018	7	20	12	35	0.93333333
2018	7	20	12	40	0.93333333
2018	7	20	12	45	0.93333333
2018	7	20	12	50	0.66666667
2018	7	20	12	55	0.66666667
2018	7	20	13	0	0.66666667

Annex III-G: Conduit Data

Table 20: Conduit Data

ID_link	From_ Node	To_ Node	Length (m)	Material	Shape_ Xsection	Diameter (mm)
06Q187- 06Q6	06Q187	06Q6	11.3	CONCRETE	EGG	450
06Q5- 06L68	06Q5	06L68	35.55	CONCRETE	EGG	450
06L72- 06L73	06L72	06L73	50.71	CONCRETE	EGG	450
06Q6- 06Q7	06Q6	06Q7	34.99	CONCRETE	EGG	450
06L71- 06L72	06L71	06L72	50.54	CONCRETE	EGG	450
06L68- 06L66	06L68	06L66	40.96	CONCRETE	EGG	450
06L67- 06L68	06L67	06L68	39.89	BRICK	CIRCULAR	250

06L70-06L71	06L70	06L71	42.93	CONCRETE	EGG	450
06L73-06L74	06L73	06L74	50.23	CONCRETE	EGG	450
06Q7-06L71	06Q7	06L71	49.44	CONCRETE	EGG	450
06L94-06L95	06L94	06L95	28.66	CONCRETE	EGG	600
06L30-06L90	06L30	06L90	28.83	CONCRETE	EGG	600
06L58-06L59	06L58	06L59	40.03	CONCRETE	EGG	450
06L64-06L30	06L64	06L30	45.67	CONCRETE	EGG	600
06L57-06L58	06L57	06L58	39.18	CONCRETE	EGG	450
06L121-06L57	06L121	06L57	32.92	CONCRETE	EGG	450
06L92-06L94	06L92	06L94	134.35	CONCRETE	EGG	600
06L62-06L63	06L62	06L63	45.77	CONCRETE	EGG	600
06L95-06L98	06L95	06L98	40.28	CONCRETE	EGG	600
06L69-06L70	06L69	06L70	42.33	CONCRETE	EGG	450
06L63-06L64	06L63	06L64	44.3	CONCRETE	EGG	600
06L61-06L62	06L61	06L62	43.19	CONCRETE	EGG	525
06L66-06L65	06L66	06L65	42.22	CONCRETE	EGG	450
06L59-06L60	06L59	06L60	43.8	CONCRETE	EGG	525
06L90-06L91	06L90	06L91	12.38	CONCRETE	EGG	600
06L60-06L61	06L60	06L61	44.17	CONCRETE	EGG	525

06L69-06L62	06L69	06L62	42.6	CONCRETE	EGG	450
06L65-06L59	06L65	06L59	42.07	CONCRETE	EGG	450
06L56-06L121	06L56	06L121	7.28	CONCRETE	EGG	450
06L91-06L92	06L91	06L92	45.08	CONCRETE	EGG	600
06L32-06L30	06L32	06L30	84.08	CONCRETE	EGG	450
06M58-06M57	06M58	06M57	34.92	CONCRETE	EGG	375
06R42-06R45	06R42	06R45	69.54	CONCRETE	EGG	450
06L74-06L32	06L74	06L32	50.31	CONCRETE	EGG	450
06L100-06L46	06L100	06L46	119.28	CONCRETE	EGG	1050
06R48-06Q18	06R48	06Q18	73.46	CONCRETE	EGG	450
06R45-06R48	06R45	06R48	127.71	CONCRETE	EGG	450
06R6-06R5	06R6	06R5	40.51	CONCRETE	EGG	600
06M56-06M57	06M56	06M57	36.9	CONCRETE	EGG	450
06Q18-06L100	06Q18	06L100	117.53	CONCRETE	EGG	900
06M61-06L51	06M61	06L51	33.33	CONCRETE	EGG	600
06L51-06L100	06L51	06L100	48.6	CONCRETE	EGG	600
06R40-06R42	06R40	06R42	43.01	CONCRETE	EGG	375
06M54-06M55	06M54	06M55	38.76	CONCRETE	EGG	450
06R5-06R4	06R5	06R4	40.41	CONCRETE	EGG	600

06M55-06M63	06M55	06M63	27.64	CONCRETE	EGG	450
06Q136-06Q18	06Q136	06Q18	36.18	BRICK	CIRCULAR	300
06R2-06R1	06R2	06R1	39.75	CONCRETE	EGG	600
06R1-06M61	06R1	06M61	40.06	CONCRETE	EGG	600
06R4-06R3	06R4	06R3	39.89	CONCRETE	EGG	600
06R3-06R2	06R3	06R2	40.18	CONCRETE	EGG	600
06M60-06M63	06M60	06M63	41.36	BRICK	CIRCULAR	250
06R103-06R40	06R103	06R40	44.73	CONCRETE	EGG	375
06R22-06R7	06R22	06R7	36	CONCRETE	EGG	375
06R25-06R103	06R25	06R103	43.06	CONCRETE	EGG	375
06R7-06R6	06R7	06R6	17.81	CONCRETE	EGG	600
06L46-06L45	06L46	06L45	34.8	CONCRETE	EGG	1050
06M55-06M39	06M55	06M39	119.73	CONCRETE	EGG	450
06M64-06M58	06M64	06M58	61.1	BRICK	CIRCULAR	300
06L31-06L46	06L31	06L46	23.59	CONCRETE	EGG	750
06M57-06M41	06M57	06M41	99.55	CONCRETE	EGG	450
06L44-06M5	06L44	06M5	172.19	CONCRETE	EGG	1050
06M39-06M38	06M39	06M38	48.05	CONCRETE	EGG	525
06M38-06L44	06M38	06L44	21.51	CONCRETE	EGG	525

06M46-06M44	06M46	06M44	47.06	CONCRETE	EGG	525
06L30-06L31	06L30	06L31	5.71	CONCRETE	EGG	750
06M40-06M39	06M40	06M39	34.35	CONCRETE	EGG	450
06M47-06M46	06M47	06M46	22.85	BRICK	CIRCULAR	300
06M42-06M41	06M42	06M41	6.65	CONCRETE	EGG	450
06M41-06M40	06M41	06M40	33.58	CONCRETE	EGG	450
06M5-06M4	06M5	06M4	16.46	CONCRETE	EGG	1050
06M44-06M42	06M44	06M42	39.33	CONCRETE	EGG	525
06L45-06L44	06L45	06L44	30.49	CONCRETE	EGG	1050
06M63-06M56	06M63	06M56	9.24	CONCRETE	EGG	450

Annex III-H: Junction Data

Table 21: Junction Data

ID	Invert elevation (m.a.s.l.)	Manhole Depth (m)	X-coordinate	Y-coordinate
06L66	149.99	3.92	410751.143	5653515.603
06L58	146.51	3.64	410754.133	5653587.807
06Q5	152.93	3.2	410762.447	5653465.788
06Q187	153.52	3.43	410783.003	5653443.446
06L67	152.08	2.97	410693.202	5653495.299
06L57	146.74	3.53	410714.954	5653587.567
06L59	146.31	3.67	410794.165	5653588.077
06L121	146.9	3.45	410682.031	5653587.373
06L65	148.16	3.68	410772.759	5653551.87
06Q6	153.26	3.21	410793.561	5653447.463
06L56	146.94	3.44	410674.751	5653587.319

06Q136	159.52	3.08	411066.916	5653334.17
06R48	159.77	2.46	411158.821	5653348.642
06R6	154.62	2.27	411385.275	5653398.073
06R45	162.81	2.65	411271.951	5653296.042
06R22	156.48	3.82	411398.593	5653353.354
06R25	167.14	3.68	411424.404	5653223.89
06R103	166.95	3.83	411381.836	5653230.385
06R5	153.97	3.12	411346.506	5653409.832
06R40	166.81	3.51	411337.66	5653237.286
06R4	153.36	3.4	411307.871	5653421.678
06R3	152.77	3.22	411269.541	5653432.714
06R42	166.53	3.15	411298.344	5653251.171
06L63	143.9	3.61	410971.09	5653588.9
06L70	149.05	3.61	410891.7	5653510.705
06L60	145.95	3.62	410837.963	5653588.246
06L62	145.15	3.54	410925.32	5653588.689
06L61	145.54	3.6	410882.134	5653588.452
06L74	147.57	3.61	411023.292	5653499.452
06Q7	152.46	3.25	410826.885	5653458.122
06L64	142.75	3.63	411015.385	5653589.067
06L73	148.8	3.53	410973.662	5653491.771
06L72	150.01	3.44	410923.769	5653482.701
06L69	147.1	3.47	410908.437	5653549.579
06M38	137.84	4.18	411131.442	5653646.469
06L100	147.36	4.69	411077.519	5653480.242
06M60	148.86	2.49	411209.01	5653485.561
06L90	140.27	3.69	411056.852	5653617.754
06L92	137.39	2.84	411043.358	5653672.663
06L91	139.59	3.62	411050.091	5653628.097
06M63	144.79	3.43	411203.43	5653526.283
06L31	141.32	3.96	411066.705	5653590.008
06M54	145.63	3.72	411137.812	5653516.148

06R1	151.53	3.04	411193.029	5653455.845
06M40	138.62	3.93	411213.245	5653636.638
06L51	150.41	2.75	411122.814	5653477.191
06M61	150.92	3.01	411154.711	5653467.514
06M56	144.47	3.91	411212.564	5653527.709
06L94	131.56	3.59	411023.687	5653805.54
06L95	131.21	3.8	411019.856	5653833.935
06M5	132.05	4.04	411142.711	5653806.202
06M46	139.13	3.64	411338.924	5653621.248
06M64	144.45	2.2	411343.917	5653548.019
06M44	138.99	3.87	411292.223	5653627.064
06M58	144.16	3.22	411283.534	5653538.665
06M42	138.82	3.76	411253.174	5653631.765
06M47	139.27	3.31	411361.775	5653620.952
06R2	152.16	2.78	411231.074	5653444.324
06L71	150.94	4.23	410874.512	5653471.393
06L68	151.31	2.45	410730.188	5653480.413
06L30	141.19	3.24	411061.043	5653589.247
06L32	145.18	3.81	411073.163	5653506.063
06Q18	155.83	2.61	411087.399	5653363.988
06M55	143.83	2.79	411176.117	5653522.082
06R7	155.24	2.27	411400.76	5653389.28
06L46	140.49	2.46	411089.99	5653593.646
06M57	142.85	4.15	411249.015	5653533.406
06L44	137.2	4.25	411111.805	5653655.147
06M39	138.29	2.97	411179.153	5653640.806
06M41	138.79	3.77	411246.574	5653632.548
06L45	138.75	3.81	411101.89	5653626.321
06L98	129.72	0	411014.305	5653873.824
06M4	131.58	0	411146.928	5653822.107

